IBEX RISC-V · 68 SECURITY ASSERTIONS · 9 MODULES · TARGET IEEE ETS 2027

Assertion Quality Ranker

A beginner's guide to how we measure whether a hardware security check is actually good. It covers the idea, the three metrics with real Ibex examples, what other researchers have done, what our red-team and blue-team reviews found, and the experiments still to run.

Advisor approved: SRTLC · AER · SAPC

Comparison only: CWE/CVSS · Trojan category

Review: SAPC and AER definitions need fixing

The idea Key words 1 · SRTLC 2 · AER 3 · SAPC The score What others did Red and blue team What we can claim Experiments Decisions

watched / caught · not watched / missed · test signal · security rule / score

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START HERE

The idea in five steps

Step 1

A processor

Ibex is a small open-source RISC-V processor. Its design is written as code (RTL).

Step 2

Security checks

We already have 68 *assertions*: small rules such as "user programs may not change machine registers". All 68 are proven true on Ibex (our BEC 2026 paper).

Step 3

Are the checks good?

A check can be true and still weak: it may watch only one way of breaking the rule. Nobody has a formal way to say how strong each check is.

Step 4

Give each a score

Three metrics, computed by the JasperGold tool on the clean design, give every assertion a quality score called **AQS**.

Step 5

Test the score

We hide 39 hardware Trojans in Ibex. If high-scoring assertions catch more Trojans, the score is meaningful.

> **Why it matters:** a chip designer can then say "this set of checks covers our security logic well", or see exactly which checks are weak and what to add.

VOCABULARY

Key words in plain language

RTL

The code that describes a chip's circuits, written in SystemVerilog.

Assertion

A rule written in code that must always hold, e.g. "in user mode, never write `mstatus`". If it is broken, the assertion "fires".

JasperGold

A Cadence tool that checks rules mathematically, for every possible input, instead of running test cases.

FPV

Formal Property Verification. The JasperGold mode that asks "can any input sequence break this rule?"

SPV

Security Path Verification. The JasperGold mode that asks "can data travel from here to there through the circuit?"

Proven / counterexample

Proven means the rule can never break. A counterexample (CEX) is a concrete step-by-step case where it does.

Cone of influence (COI)

All the circuit logic that can affect an assertion. It is the part of the design the assertion "sees".

Hardware Trojan

A small malicious change hidden in a chip design, e.g. a backdoor that grants machine

privileges.

Non-vacuous

The assertion's "if" part can actually happen, so it is not passing only because it never gets tested.

`mstatus`

An Ibex machine-mode control register. Changing it from user mode would be a privilege attack.

Metric 1 · Security RTL Coverage (SRTLC)

`JasperGold COI report · 0 new runs`

How much of the security logic do our checks look at?

Everyday picture

A bank has 40 important doors. The security cameras (assertions) can see 30 of them. Coverage is 30 / 40 = 75%. The other 10 doors are unwatched.

1. Which code lines are "security logic"? Those that drive signals named in the expert NS31A security properties. We don't guess them.
2. JasperGold reports each assertion's cone of influence: the lines it can see.
3. Count the security lines that at least one assertion sees.

$$\text{SRTLC} = \frac{\text{security lines seen by some assertion}}{\text{all security lines in the module}} \times 100\%$$

L101	L108	L115	L122	L129	L136	L143	L150	L157	L164
L171	L178	L185	L192	L199	L206	L213	L220	L227	L234
L241	L248	L255	L262	L269	L276	L283	L290	L297	L304
L311	L318	L325	L332	L339	L346	L353	L360	L367	L374

75% 30 of 40 security lines seen (example, CSR module)

Each box is one line of security code. Red boxes are lines no assertion can see, so a Trojan placed there could go unnoticed.

> **Review note:** COI coverage can overestimate what an assertion really checks. Both review teams suggest also reporting *proof-core* coverage, a stricter JasperGold option.

Metric 2 · Assertion Evasion Resistance (AER)

`JasperGold FPV + test signal · 10 runs`

Can someone break the security rule without this check noticing?

Everyday picture: testing a smoke alarm

The goal is that the kitchen never burns unnoticed. The smoke alarm is our check. To test it properly, we let a helper start a small fire anywhere, at any time, in any way. If the kitchen can burn while the alarm stays quiet, the alarm is weak. If every fire, in every place, sets it off, the alarm is strong.

In AER, JasperGold plays the helper, and it tries **every** possibility, not just a few.

The three players (real Ibex example)

`THE SECURITY RULE (P)`

What must never happen. This is the goal, written once.

```
In user mode, mstatus
must never change.
```

`THE ASSERTION UNDER TEST (A)`

One of our 68 checks. AER scores this one.

```
UserMode && CsrWriteOp
  |-> !WriteEn_mstatus
```

"If user mode AND a CSR write instruction, then the mstatus write-enable is 0."

`THE TEST SIGNAL`

An extra input JasperGold controls freely. It can flip the `mstatus` write-enable wire on any clock cycle.

```
we_seen = we_real XOR test_sig
```

JasperGold then asks one question: **is there any way to set the test signal so that rule P breaks while assertion A stays silent?**

Weak assertion (A) | Stronger assertion

Clock cycle	Mode	CSR write instruction?	Test signal flips write-enable?	mstatus changed?	Rule P	Assertion A
1	user	no	no	no	OK	silent
2	user	no	**yes**	**yes**	BROKEN	silent
3	user	no	no	no	OK	silent

Result: counterexample, AER = 0

In cycle 2 the write-enable is flipped directly, without any CSR instruction. The "if" part of A (`CsrWriteOp`) is false, so A never checks anything, and mstatus changes in user mode unnoticed. JasperGold shows this exact trace, which tells us precisely why A is weak: **it watches the instruction, not the register.**

```
AER = 1  if no test-signal pattern breaks rule P while A stays silent
AER = 0  if JasperGold finds one (and shows the exact trace)
```

> **Important:** rule P and assertion A must be different. If we used A as its own rule, A could never "break silently" and every assertion would score 1. The review teams suggest taking P from the original NS31A expert property. This is a decision for us (see decisions). The table above is an illustration; real AER values come from the JasperGold runs, which are not done yet.

Metric 3 · Security Assertion Path Coverage (SAPC)

JasperGold SPV · 10 runs

Are there roads into the security register that no alarm is watching?

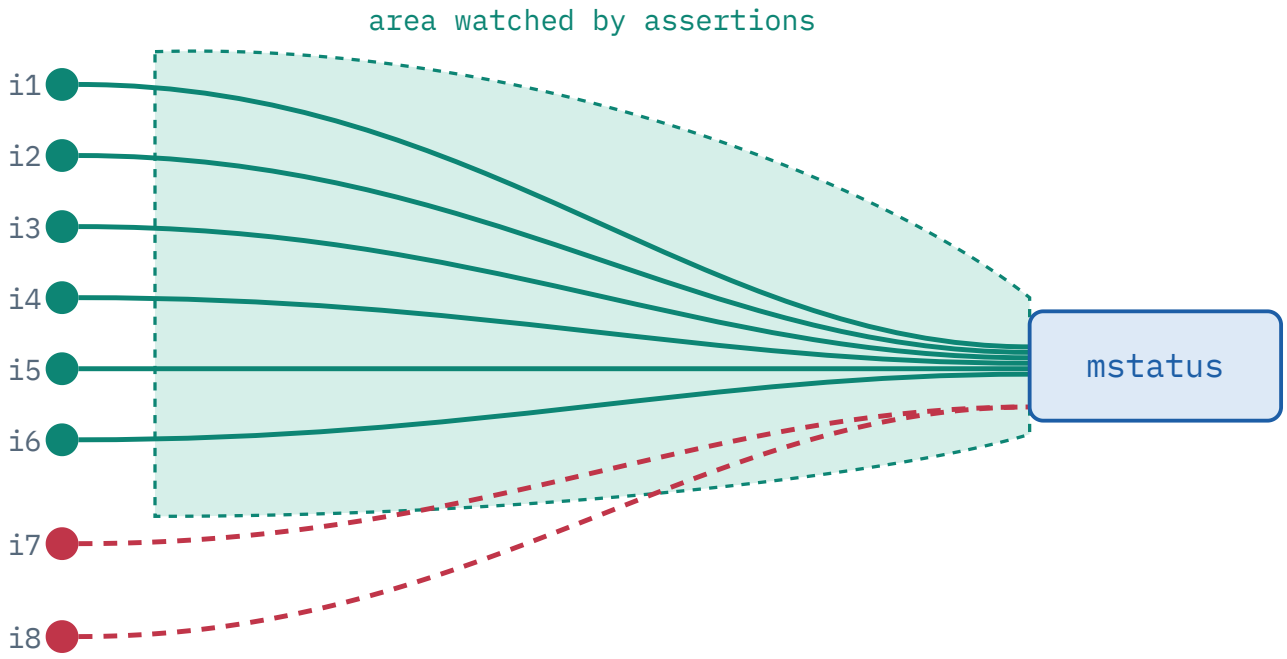
Everyday picture

Eight roads lead to the vault (`mstatus`). For each road we ask whether a car on it would trigger any alarm. Six roads would, two would not. SAPC = 6 / 8 = 75%.

1. JasperGold SPV lists every path from a chip input to a security register.
2. For each path, check whether a change travelling along it reaches some assertion.
3. Paths that reach no assertion are blind spots.

$$\text{SAPC} = \frac{\text{paths into security registers watched by an assertion}}{\text{all paths into security registers}} \times 100\%$$

> **How it differs from AER:** AER only flips the one wire the test signal is attached to. SAPC looks at *all* roads, including ones AER never touched.



75% 6 of 8 roads watched (example)

Dots on the left are chip inputs. Green roads are watched; the two dashed red roads reach `mstatus` with no alarm.

Problem the review found, and the fix

If "watched" just means "the road lies inside the assertion's cone of influence", every assertion that looks at `mstatus` automatically watches *all* roads into it. The number would be almost always 100% or 0% and would just repeat SRTLC.

Fix: for each road, ask SPV whether a change sent down it actually travels through `mstatus` all the way to the assertion's failure output. This follows the method of our co-author's earlier work (Eslami, Ghasempouri, Pagliarini, 2022).

PUTTING IT TOGETHER

The Assertion Quality Score (AQS)

Each metric gives a number between 0 and 1 for every assertion. AQS is their average:

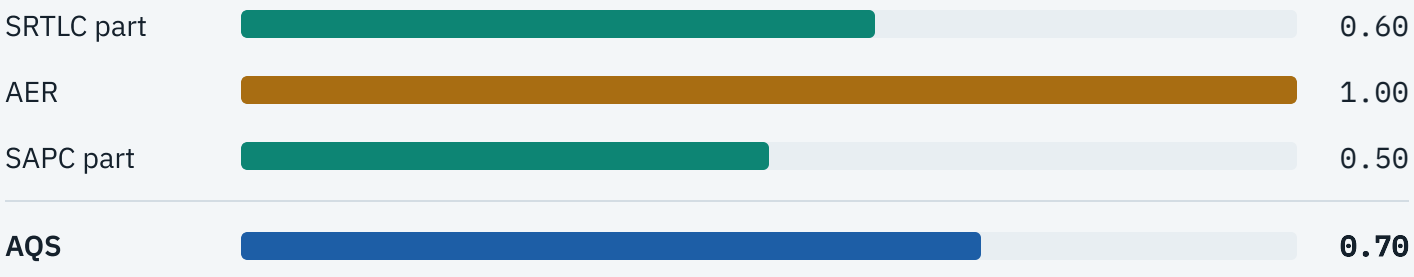
```
AQS = (SRTLC part + AER + SAPC part) / 3
```

How we check the score works: after scoring, we run the same 68 assertions against 39 Trojan-infected versions of Ibex and count how many Trojans each one catches. If assertions with higher AQS catch more Trojans, the score predicts real protection. Statistically this is a *correlation*; the reviews recommend Spearman's rank correlation with confidence intervals.

Beside the score: two descriptive tables

Table A: which known weakness each assertion protects against (CWE number and its CVSS severity, following NIST CSWP 45).

Table B: which Trojan types each assertion targets (Availability, Covert, Denial of Service, Integrity, Leak, Privilege). These describe the assertions; they are not part of the score.



> Example numbers for one assertion, not measured results.

LITERATURE · 72 PAPERS SURVEYED

What other researchers have already done

Our literature survey found work close to every one of our metrics. That is normal. It tells us which ideas to credit and where our contribution really lies. The papers below are the most important, grouped by the question they answer.

Closest to AER

"Does the check notice when the design is changed?"

Chockler, Kupferman, Vardi et al., 2001

Change the design slightly (mutation) and see whether the property notices. The classic idea behind AER.

Kupferman, CONCUR 2006

"Sanity checks" after a proof: vacuity and coverage. The framing for our whole paper.

Krishnamurthy et al. 2026 (arXiv 2606.21451); HierSVA 2026 (arXiv 2606.13706)

Score LLM-written assertions by how many injected faults they catch.

For us: AER differs because it checks *all* flip patterns in one proof, against a security rule.

Chuah, Appold, Leinmüller: MBMV 2024; RISC-V Summit Europe 2025; ISQED 2026 (ISQED page)

Formal Trojan detection on a RISC-V core with Jasper. The red team found the same "XOR free input" trick in their 2025 abstract.

For us: cite them as the origin of the mechanism; our addition is a score per assertion.

Closest to SAPC

"Can a change travel to the check?"

Eslami, Ghasempouri, Pagliarini, ISQED 2022 (arXiv 2201.01130)

Uses JasperGold SPV to count how many circuit points can reach each assertion ("Security Coverage") and ranks assertions by it. Scores were about 5–7% because they divide by the whole circuit.

For us: our own group's work. SAPC narrows it to security registers and tests it against Trojans.

Eslami et al., SCARF, IEEE Trans. Computers 2024 (arXiv 2402.12162)

Journal version of the same per-assertion security coverage.

Meza & Kastner, 2023 (arXiv 2304.08263)

Coverage metrics based on how information flows through hardware.

Closest to SRTLC

"How much of the design does the check see?"

Closest to AQS ranking and our validation

"Which checks are best, and do they catch Trojans?"

JasperGold coverage app (Cadence)

Built-in cone-of-influence and proof-core coverage. Cadence's own material says COI overestimates.

For us: SRTLC's new part is counting only expert-flagged security lines. Also report proof-core.

LAsset (DATE 2026); SAIF (VTS 2021)

Find which signals are security "assets". This decides what counts as security logic.

Ayalasomayajula et al., HOST 2024; Assert-O 2025; Hertz et al., ASP-DAC 2019

Earlier ways to rank assertions (by severity, by combined merit scores).

For us: AQS ranks by formal evidence, not severity labels.

Imtiaz, Reinsalu, Ghasempouri, ISCAS 2025 (arXiv 2502.10194)

Our own earlier paper: Ibex assertions tested on Trojans with the TDER and TPI measures.

For us: reuse and cite TDER/TPI; don't present them as new.

Inozemtseva & Holmes, ICSE 2014

In software testing, coverage correlates only weakly with bug-finding once test size is controlled.

For us: a warning. Control for assertion size when correlating AQS with detection.

Trojan benchmarks

"Where do test Trojans come from?"

TrustHub (Shakya et al., 2017)

The standard public Trojan benchmark and its categories.

Krieg, ICCAD 2023

Found only 3 of 83 TrustHub designs are real Trojans.

For us: say "TrustHub categories, new Ibex code", not "TrustHub Trojans".

Damljanovic et al., DDECS 2021; RISC-V-Dilemma-1, 2025

Earlier processor Trojan benchmarks.

For us: don't claim the first RISC-V Trojan benchmark.

> Full list with links: `docs/research/literature_survey.md`. About 15 entries are marked unverified and must be checked before citing.

NOVELTY REVIEW

Red team and blue team

Red team: the hostile reviewer

Searched about 30 sources and tried to destroy every novelty claim, the way a tough ETS reviewer would.

Blue team: the defence lawyer

Searched about 25 sources and built the strongest *honest* case, plus the wording and experiments that survive the red team.

Item	Red team says	Blue team says	In plain words
SRTLC	Partly done before; COI coverage is a standard tool feature.	Defensible: the expert-defined security denominator is new.	`keep` Call it an adaptation. Add proof-core coverage.
AER	Mechanism already published (Chuah 2025); rule P undefined.	A formal relative of mutation coverage, but new as a per-assertion security score.	`fix` Define P separately give a graded score instead of 0/1.
SAPC	Done by our co-author (2022), and may simply repeat SRTLC.	Valid if "watched" means a real path to the assertion's output.	`must fix` Redefine as functional paths; credit Eslami 2022.
AQS ranking	Assertion ranking exists already.	New: combines behaviour (FPV) and paths (SPV) in one score.	`keep` Claim the combination, never "first"
Score predicts Trojans	Looks new, but small data and our own Trojans.	The strongest claim, if done carefully.	`fix` Rank statistics, baselines, held-out Trojans.

The four problems to solve

MUST FIX · SAPC

SAPC may count nothing new

Measured with the cone of influence, any assertion that looks at a register "watches" every road into it.

Fix: count only roads where SPV shows a change really reaches the assertion's failure output.

MAJOR · AER

Which rule is P?

If P is the assertion itself, AER is always 1. A fully free test signal may also make almost every assertion score 0.

Fix: take P from the NS31A expert property; score how many flip points or how much tampering the assertion catches (graded).

MAJOR · VALIDATION

The Trojan test could look rigged

We wrote the Trojans ourselves, the sample is small, and refinement would be tested on the same Trojans that guided it.

Fix: freeze the Trojans first, keep some aside for the final test, compare with simple baselines.

MAJOR · POSITIONING

No "first" claims

Every single metric has a close ancestor in the literature.

Fix: credit Chuah and Eslami openly and claim the combination plus validation.

OUR HONEST CONTRIBUTION

What we can claim

Proposed novelty sentence (blue team): AQS is a per-assertion security quality score computed by formal analysis on Trojan-free RTL, combining coverage of expert-defined security signals (SRTLC), FPV-based evasion resistance (AER) and SPV-based functional-path coverage (SAPC). On 68 proven Ibex assertions across 9 security modules, AQS computed before any Trojan exists correlates with detection on an independent 39-Trojan benchmark, and AQS-guided refinement improves detection on held-out Trojans.

- **Safe to say:** "to our knowledge, the first to combine FPV evasion and SPV path evidence in one per-assertion score".
- **Our cleanest new insight, if the data shows it:** an assertion can be behaviourally strong (AER high) and still leave roads unwatched (SAPC low).

- **Never say:** "first metric", "first RISC-V Trojan benchmark", or that AQS "proves" detection. Say "correlates with".

NEXT STEP

Five stages and the experiments to run

Stage 1 · AVS

Validation

Done in BEC 2026: 68 assertions proven and non-vacuous.

`0 new runs`

Stage 2 · AQRS

Ranking

SRTLC, AER, SAPC on the clean design; AQS and both tables.

`10 FPV + 10 SPV`

Stage 3 · TES

Trojan test

Assertions on 39 Trojan versions; does AQS predict detection?

`39 FPV`

Stage 4 · ARS

Refinement

Add template assertions for the gaps found. Python, no AI.

`0 runs`

Stage 5 · RAVS

Re-test

Re-run on held-out Trojans and measure the gain.

`39 FPV`

ID	Experiment	Stage · tool	Runs	What it produces	Status
E1	SRTLC per module and per assertion (COI, plus proof-core)	AQRS · Python	0	Bar chart per module; list of unwatched security lines	core
E2	AER with test-signal wrappers, one per bind file	AQRS · FPV	10	AER per assertion; the evasion trace for each weak one	core
E3	SAPC functional paths into each security register	AQRS · SPV	10	Path coverage per module; list of unwatched roads	core
E4	AQS ranking of all 68 assertions	AQRS · Python	0	Ranked table, best and weakest per module	core
E5	CWE/CVSS and Trojan-category tables	AQRS · Python	0	Tables A and B	core
E6	Run unchanged assertions on the 39 Trojan designs	TES · FPV	39	Detection heat map; TDER, WTDR, AAD	Trojans
E7	Does AQS predict detection? (Spearman with confidence intervals)	TES · Python	0	Scatter plot; correlation with and without size control	core
E8	Baselines: cone size, number of signals, random ranking	TES · Python	0	Does AQS beat simple predictors?	proposed
E9	Ablation: drop one metric at a time; random weights	TES · Python	0	Is each metric needed? Is the ranking stable?	proposed
E10	Head-to-head: AER vs mutation kill rate; SAPC vs Eslami coverage	AQRS · FPV/SPV	≈10	One worked case the older metric misses	proposed
E11	Refine assertions from the gap report	ARS · Python	0	New template assertions per gap	core
E12	Re-test refined assertions on held-out Trojans	RAVS · FPV	39	Before/after detection on Trojans not used for refinement	Trojans

BEFORE EXPERIMENTS START

Decisions for Sharjeel and the advisor

1. **SAPC:** redefine "watched" as a functional SPV path to the assertion's failure output?
2. **AER:** take rule P from the NS31A expert property? Graded score or 0/1?
3. **Statistics:** Spearman with confidence intervals, and keep some Trojans aside for the final re-test?
4. **Citations:** how do the BEC 2026 paper (68 assertions, 9 modules) and the ISCAS 2025 paper (5 modules) relate?

Sharjeel Imtiaz, Uljana Reinsalu, Tara Ghasempouri · Tallinn University of Technology · Estonian Research Council PSG837

Input assertions from *AutoTrans* (BEC 2026). All numbers on this page are illustrative; no Stage 2–5 results exist yet. Full reports: `docs/research/`.